

NOTE: This document was made for Professor Jacklitsch’s 2D Game Development class at Kutztown University for 10/06/26

Deep underground become a spider, a collector of sorts. An artisan in the ways of the web, taking inspiration from the old world. In need of a crystal’s natural oil to improve your webs, embark on an adventure within the tunnels. In this dark fantasy side scroller with exploration and curiosity comes danger and the chase. Collect the crystals, jump the platforms, avoid the jaws of the Big Fish O’ Frog. Can you evade evolution or get caught in its web?



Table of Contents

Table of Contents 2
1. Terms 3
2. Game Attributes 4
3. Overview 5
4. Story 5
5. Gameplay 6
a. Player Objective 6
b. Enemy Behavior 6
c. Web Ability 7
d. Crowd Control Mechanic 7
6. Artstyle 7
a. Style Guideline 7
b. Mood Board 8
c. Example Artwork 8
7. Music 10
a. Style Guideline 10
b. Run Music 10
c. Main Menu Music 11
8. UI 11
a. Style Guideline 11
b. HUD 11
c. Menus 11
9. Technical Elements 12
a. Game Engine 12
b. Programming Languages, Libraries, and Frameworks 12
c. Key Technical approaches 12
10. External Assets 13
a. Music 13
b. Artwork 13
11. Citations 14

1. Terms

The Spider	The spider is the character the player will control.
Big Fish O’ Frog	Big Fish O’ Frog is the enemy the player will encounter during a run.
Ceilum Crystal	The Ceilum Crystal is the gem found at the beginning of the run.
Run	The entirety of one session, from load in to Win/Lose condition.

2. Game Attributes

Genre	Platformer
Genre-Adjacent Tags	Side-Scroller
Gamemodes	Single Player
Theme	Fantasy
Dimension	2D
Perspective	Side View
Unique Selling Point	Graphics and Visuals
ESRB Rating	E
Target Audience	Casual Gamers
Platforms	PC, Switch, Mobile, Console
Inspiration	Super Meat Boy, Sonic the Hedgehog, Hunt the Night, Castlevania

3. Overview

Cave Spider: Origins is a game in which the player takes a valuable crystal and must outrun the main enemy of the game, Big Fish O’ Frog. Using the rock platforms the player will have to climb to escape. The game is based around a countdown mechanic where the player must survive for the allotted time in order to win. They will get increasingly difficult between each level. The game is designed to be simple to play as to assist in ease of entertainment so the player feels more likely to pick up the game and play a few runs.

---

4. Story

Deep beneath a long forgotten planet, crumbling to pieces in its civilizations absence, the underground flourishes in the silence and undisturbed soil. Creatures once inferior to the giants above mutate and evolve into more intelligent, more destructive predators.

In the cold, damp ruins of old coal shafts The Spider becomes more sentient, more curious.

The Spider is a collector of many things, as many things can be used for making the most excellent webs. The best finds are that of the shiny Ceilum Crystal. The most beautiful, soft glowing gemstone the dirt has ever kept secret. With the giants gone, no one is here to mine the resource. All the fragments left in the planet's layers, every one off test piece in a laboratory bin, free for the taking.

This crystal is like no other, it can be broken down in many ways. Its essence can be modified into oils for protecting webbing, silks to make stronger webs, and even bait for the most delectable of all spider entrees, The Moth.

There stands one obstacle in The Spider’s way, Big Fish O’ Frog. Many species evolved beyond their sphere of normal behavior and mutated

into a new kind of Beast of Prey. Big Fish O' Frog has incredibly strong skin allowing him to effortlessly break through the thick ground, he is fast and can do something, maybe screen shake.

Big Fish O' Frog wants those crystals for making his scales shiny gold instead of an angry red. He will chase anyone who dares steal his possessions. There is one hope because he is a fish, can The Spider outrun Big Fish O' Frog until he's tucked out? Or will The Spider spin his last web?

---

5. Gameplay

a. Player Objective

The player's objective is to survive long enough without being caught by Big Fish O' Frog. The player starts out in the main menu and must press start to enter level one. In level one the player's goal is to easily skip over a few platforms and retrieve a crystal but this will cause the enemy to spawn! The player must run and enter into level two, starting the game run. As soon as you enter a run, Big Fish O' Frog begins a chase sequence with the player immediately. The player must parkour over the stone platforms within the tunnels with little error to evade Bid Fish O' Frog. Each game run starts a timer that counts down till the end of a run. The player is able to collect loose webbing from abandoned webs, allowing the player to shoot at Big Fish O' Frog and the ability to create more distance between the two entities with the crowd control mechanic.

b. Enemy Behavior

Big Fish O' Frog will follow the player the moment they come into ownership of the Ceilum Crystal, in level one. He will continue to chase The Spider through to the following level. This is the run sequence level which is set within underground tunnels. While chasing the player his speed will gradually increase making it difficult to not use the web mechanic. As the player shoots the web at Big Fish O' Frog, this will cause him to slow down

as he is temporarily impaired. Once the cool down wears off Big Fish O' Frog will speed up to be behind the player once more.

c. Web Ability

While running from the enemy the player will have the ability to collect webbing hanging around on the stone platforms. This will allow them to shoot at the enemy to slow him down, gaining more distance for a short time. Allowing the player to recoup while progressing through the level geometry.

d. Crowd Control Mechanic

This mechanic uses the web ability of the player shooting the web, to slow down the enemy as he follows the player. Once the player's web hits Big Fish O' Frog it will cause him to freeze for a short moment before accelerating up behind the character again. The enemy will have experienced a knock back in speed due to the web attack. He will still continue to gain speed to catch up to the player again, just with the speed retention of his original velocity.

6. Artstyle

a. Style Guideline

The artstyle for Game Name is one of a dark, gothic, desaturated aesthetic. Simple in color and immersive in atmosphere. The environment of unsaturated purple hues creates a cold climate in the background of the game. The only true warmth from the environment comes from the sandy brown stone littering the landscape.

The characters have two distinct silhouettes each representing their personalities and goals. The Spider is a dark purple, almost black in color, as to stand out from the murky scenery. It also has more curves and rounded edges suggesting the softness, and approachability of our protagonist. Big Fish O' Frog is a fish/ frog hybrid animal shown in his size

and deep red color. He is also curved but more so in the way to express power and consumption.

b. Mood Board



c. Example Artwork

i. Character

7

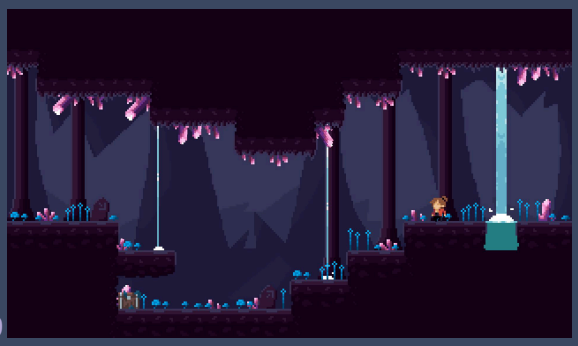


8

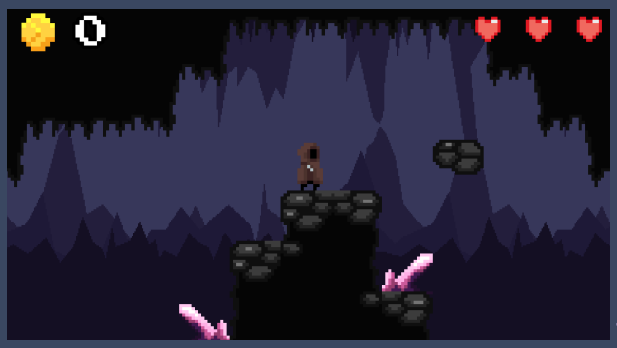


The spider should be a simple pixel design with minimal shading compared to the background to contrast the environment.

ii. Background



9



10

The environment should stick to moody purples to show a connection with The Spider. It should be more of a chunky 2d pixel style to contrast the player and enemy characters.

iii. Enemy



11



12

The enemy should be large and exaggerated in size and attributes compared to The Spider

iv. Crystal



13



14

The crystal should look enough like a gem again matching the 3 value min shading conventions. It should be yellow to contrast the player.

---

7. Music

a. Style Guideline

All music used in the game should follow the same aesthetic of the 8-bit chiptune sound. The overarching theme should be to the effect of Baroque progressions. It should be fast-paced with a clear melodic chord, exciting high energy accents, and have an overall eerie and spooky vibe.

b. Run Music

Music used for the runs will be slightly louder than the main menu as to add distraction in the levels environment. It should be intricate and a prominent take away from the game's experience. The mood and tone should be one of tension and anxiety, invoking that spooky theme embedded within the games core design.

c. Main Menu Music

Music in the Main Menu should be loud enough that it is evident and heard but not so much as it becomes gaudy or overbearing. The tone for this motif should be calmer than the overall music while still fitting into the upbeat nature of the game. The aesthetic will sound more like a buoyant ambient or lo-fi retro game theme, still following the chiptune style.

8. UI

a. Style Guideline

The style for the UI again stays consistent with the pixel art visual style. It should remain on the simpler side while still upholding the concept of a gloomy and dreary atmosphere. It will be distinguished by its bold outlines and desaturated colors. Along with slightly more rendered graphics inspired by the abandoned city.

b. HUD

i. Web Counter HUD

1. In the top right hand corner of the screen the Web Ability UI should be displayed. It will be distinguished by a black and white spider web graphic with a counter displaying the number of the webs the player has collected. This will only display the webs not yet used.

ii. Countdown Timer

1. Above the Web Counter HUD will sit the Countdown Timer. This timer will countdown from the allotted time forcing the player to survive until time's up. This should not affect game feel in any way and remain a less invasive HUD element.

c. Menus

i. Main

1. The main menu will give the player the option to exit the game, start the game, and see the credits. When the player hits the ‘exit’ game button the program will exit. When the player hits the ‘start’ button the game will enter the entry level. When the player presses the ‘credits’ button they will be taken to the credits screen.

ii. Pause

1. The pause menu will allow the player to pause the game and unpause the game, return to the main menu, or exit the game. In the run when the player presses the ‘esc’ button, the pause menu will pop up. When the player presses the ‘continue’ button the game will return to the run.

9. Technical Elements

a. Game Engine

The game engine used to create this game will be Godot.

b. Programming Languages, Libraries, and Frameworks

The programming language used will be C# as that is what is known by the development team. This will be in line with the standard Godot framework. No additional libraries or frameworks will be used beyond those inherent to the Godot Engine.

c. Key Technical approaches

i. Game Feel Mechanic

1. This mechanic uses the player’s input with the mouse to shoot projectiles. By getting the position of the mouse it will give the player a feeling of control in the game by having to aim. It will allow a big payoff when predicting the enemy movement with an immediate visual reaction.

ii. Moving platforms
    1. Farther into the run around midway to end, some of the level platforms will begin to move. This will add another immersive element into the game's sequence as well as a tense addition to the chase sequence with Big Fish O' Frog. The platforms will move back and forth, following a tempo platform design, making the timing of jumps more important.
iii. Camera Relative Control
    1. The camera will follow the player for the entire game, through every level. The enemy should remain closer to the edge until his speed starts to increase. The enemy and geometry will be able to move within and out of the frame.

10. External Assets

a. Music

i. For main: https://freesound.org/people/frankum/sounds/752104/

ii. For run: https://freesound.org/people/Seth_Makes_Sounds/sounds/681705/

iii. For SFX: https://freesound.org/people/nebulasnails/sounds/495117/

b. Artwork

i. Fonts

1. Title: https://www.dafont.com/alagard.font
2. Text: https://www.dafont.com/pearsoda.font?text=start

11. Citations

Mood Board:

1.) https://discussions.unity.com/t/spelunky-clone-open-source-2d-platformer/801098 [1]
2.) https://www.reddit.com/r/Unity2D/comments/1pcd1ix/do_unity_2d_devs_still_use_small_pixelart_packs/ [2]
3.) https://www.instagram.com/popular/2d-indie-game-pixel-art-platformer/ [3]
4.) https://www.gamedevmarket.net/asset/pixel-art-enemy-pack-5190 [4]
5.) https://www.dualshockers.com/best-spooky-pixel-art-games/ [5]
6.) https://www.vectorstock.com/royalty-free-vector/underground-cave-pixel-background-vector-49379004 [6]

Other Inspirations:

1.) https://www.istockphoto.com/vector/8-bit-pixel-spider-black-vector-pixel-art-tarantula-isolated-on-white-background-gm1481105355-508516710 [7]
2.) https://www.shutterstock.com/image-vector/spider-icon-pixel-art-draw-picture-2061053234 [8]
3.) https://ki11spika.artstation.com/projects/EVOEOK [9]
4.) https://lil-cthulhu.itch.io/pixel-art-cave-background [10]
5.) https://www.neogaf.com/threads/best-and-or-most-visually-impressive-boss-fights-in-a-2d-game.1096407/ [11]
6.) https://www.thegamer.com/hardest-metroidvania-video-games/ [12]
7.) https://en.ac-illust.com/clip-art/26010280/small-crystal-pixel-art-set-2 [13]
8.) https://en.ac-illust.com/clip-art/26068438 [14]

---

[1] Unity Discussion Board, July 2020, TwiiK
[2] Reddit, December 2025, sebbyspoons
[3] Instagram, freexelart_official
[4] GameDev Market, May 2016, froberg7
[5] DualShockers, Oct 2024, Joshua Speer
[6] VectorStock, Nov 23, ad_stock
[7] iStockPhotos,
[8] ShutterStock,
[9] ArtStation,
[10] itch.io,
[11] neogaf,
[12] TheGamer,
[13] AC Illustrator,
[14] AC Illustrator,